THE PUSHFORWARD ALONG THE LOCALIZATION OF A TOPOS IN A SLICE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The localization morphism attached to a sheaf. Same chapter, Lemma lemma-localize-topos, recorded as the context of this card. Supplies the morphism of topoi whose pushforward is computed.
- The adjunction between products with a sheaf and the internal hom. Same chapter, Lemma lemma-internal-hom-sheaf. Turns the morphism sets into values of a sheaf of right inverses.

Source reference. Original section 00XZ: sources/stacks-sites/pinned-original-source.tex, source lines 7554–7554. Navigation pointer invoked at proof line 7554; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BWQ: sources/stacks-sites/pinned-original-source.tex, source lines 7624–7624. Navigation pointer invoked at proof line 7624; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. Then the category $Sh(\mathcal{C})/\mathcal{F}$ is a topos. There is a canonical morphism of topoi*

$$j_{\mathcal{F}} : Sh(\mathcal{C})/\mathcal{F} \longrightarrow Sh(\mathcal{C})$$

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which is a localization as in Section 00XZ such that

(1) *the functor $j_{\mathcal{F}}^{-1}$ is the functor $\mathcal{H} \mapsto \mathcal{H} \times \mathcal{F}/\mathcal{F}$, and*
(2) *the functor $j_{\mathcal{F}!}$ is the forgetful functor $\mathcal{G}/\mathcal{F} \mapsto \mathcal{G}$.*

Lemma 0.2. *In the situation of Lemma 0.1, the functor $j_{\mathcal{F},*}$ is the one associates to $\varphi : \mathcal{G} \to \mathcal{F}$ the sheaf*

$$U \longmapsto \{\alpha : \mathcal{F}|_U \to \mathcal{G}|_U \text{ such that } \alpha \text{ is a right inverse to } \varphi|_U\}.$$

Proof. For any $\varphi : \mathcal{G} \to \mathcal{F}$, let us use the notation $\mathcal{G}/\mathcal{F}$ to denote the corresponding object of $Sh(\mathcal{C})/\mathcal{F}$. We have

$$(j_{\mathcal{F},*}(\mathcal{G}/\mathcal{F}))(U) = \mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\#, j_{\mathcal{F},*}(\mathcal{G}/\mathcal{F})) = \mathrm{Mor}_{Sh(\mathcal{C})/\mathcal{F}}(j_{\mathcal{F}}^{-1}h_U^\#, (\mathcal{G}/\mathcal{F})).$$

By Lemma 0.1 this set is the fiber over the element $h_U^\# \times \mathcal{F} \to \mathcal{F}$ under the map of sets

$$\mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\# \times \mathcal{F}, \mathcal{G}) \xrightarrow{\varphi\circ} \mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\# \times \mathcal{F}, \mathcal{F}).$$

By the adjunction in Lemma 0BWQ, we have

$$\begin{aligned}
\mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\# \times \mathcal{F}, \mathcal{G}) &= \mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\#, \mathcal{H}om(\mathcal{F}, \mathcal{G})) \\
&= \mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{G}|_{\mathcal{C}/U}), \\
\mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\# \times \mathcal{F}, \mathcal{F}) &= \mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\#, \mathcal{H}om(\mathcal{F}, \mathcal{F})) \\
&= \mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{F}|_{\mathcal{C}/U}),
\end{aligned}$$

and under the adjunction, the map $\varphi\circ$ becomes the map

$$\mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{G}|_{\mathcal{C}/U}) \longrightarrow \mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{F}|_{\mathcal{C}/U}), \quad \psi \longmapsto \varphi|_{\mathcal{C}/U} \circ \psi,$$

the element $h_U^\# \times \mathcal{F} \to \mathcal{F}$ becomes $\mathrm{id}_{\mathcal{F}|_{\mathcal{C}/U}}$. Therefore $(j_{\mathcal{F},*}\mathcal{G}/\mathcal{F})(U)$ is isomorphic to the fiber of $\mathrm{id}_{\mathcal{F}|_{\mathcal{C}/U}}$ under the map

$$\mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{G}|_{\mathcal{C}/U}) \xrightarrow{\varphi|_{\mathcal{C}/U}\circ} \mathrm{Mor}_{Sh(\mathcal{C}/U)}(\mathcal{F}|_{\mathcal{C}/U}, \mathcal{F}|_{\mathcal{C}/U}),$$

which is $\{\alpha : \mathcal{F}|_U \to \mathcal{G}|_U$ such that α is a right inverse to $\varphi|_U\}$ as desired. □